

Introduction to Machine Learning

Lecture 8: Classification with Confidence

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Quick Recap: *Classification*

We discussed various approaches for classification

- *linear regression with signed labels*
- *logistic regression*
- *maximum likelihood $\equiv$ minimizing cross-entropy*

At the end of the day all of these approaches find a

Linear Classifier

which is not unique!

Today's Agenda: *Support Vector Classifier*

Today, we find the linear classifier with maximal confidence

Support Vector Classifier (SVC)

In this way, we discuss the following topics

- *We define a notion of margin as a confidence representative*
- *We formulate linear classification with maximal margin*
 - ↳ *This ends us up with SVC*
- *We learn how we can solve the underlying optimization in SVC*
 - ↳ *We review the method of Lagrange multipliers*

Earlier Example: *Cat or Dog*

- *Dataset*

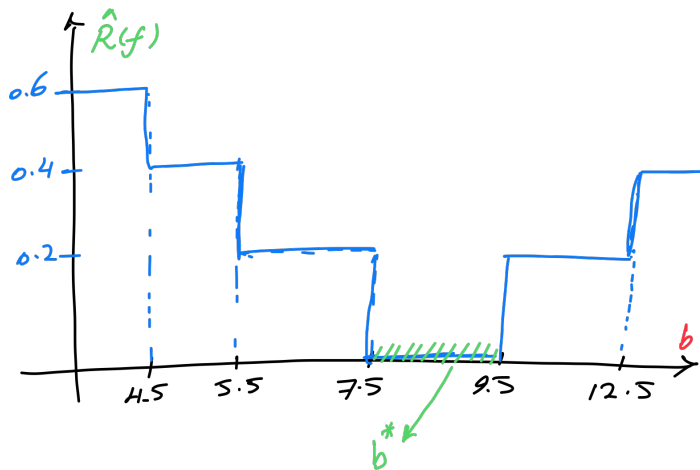
$$\mathbb{D} = \{(x, v) = (5.5, \text{cat}), (4.5, \text{cat}), (12.5, \text{dog}), (9.5, \text{dog}), (7.5, \text{cat})\}$$

- *Linear classifier*

$$y = \begin{cases} \text{cat} & x < b \\ \text{dog} & x \geq b \end{cases}$$

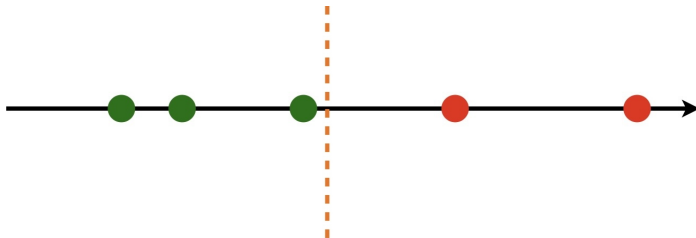
- *Learning algorithm* $\equiv$ *empirical risk minimization*

Earlier Example: Cat or Dog

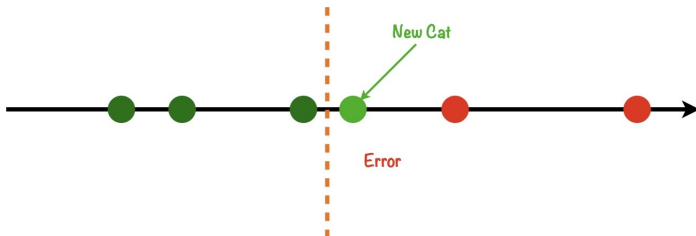


Earlier Example: Cat or Dog

This threshold is optimal

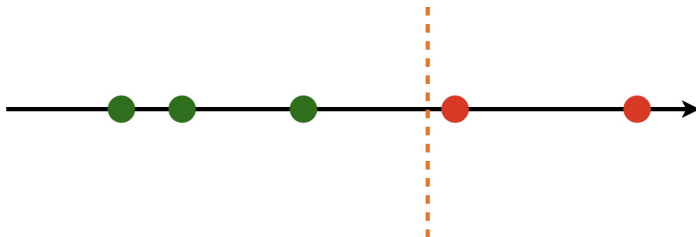


*But seems to have poor **confidence** with new samples*

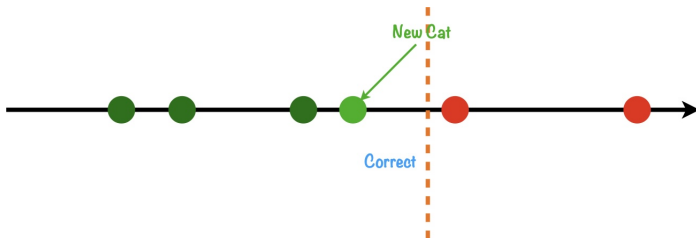


Earlier Example: Cat or Dog

Maybe we can push right

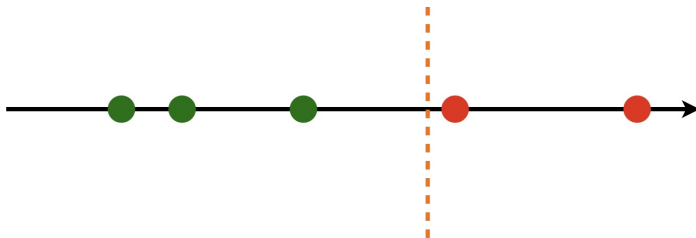


It seems to work

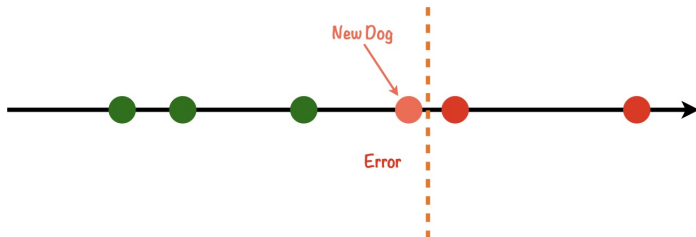


Earlier Example: Cat or Dog

But we could have a new *dog* sample

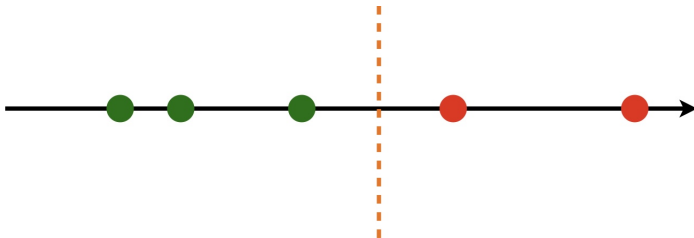


It seems not to be confident again

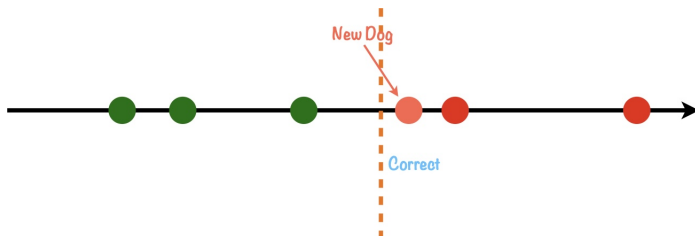


Earlier Example: Cat or Dog

Maybe we can push left



Middle point seems to be best



Confidence for Generalization

We have in general many thresholds that minimize the empirical risk

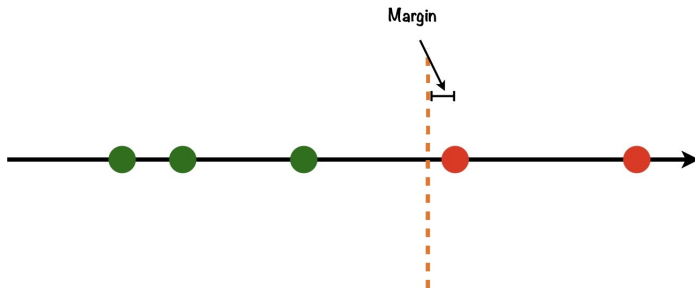
- *We could look among them for the one that gives us maximal confidence*
- *Confidence describes how well we think the choice generalizes*
 - ↳ *This threshold works perfect on the dataset*
 - ↳ *We want to minimize the chance of having error with new samples*
- *For maximal confidence, we could maximize the margin*

? *How do we define the margin?*

Defining Margins

Margin

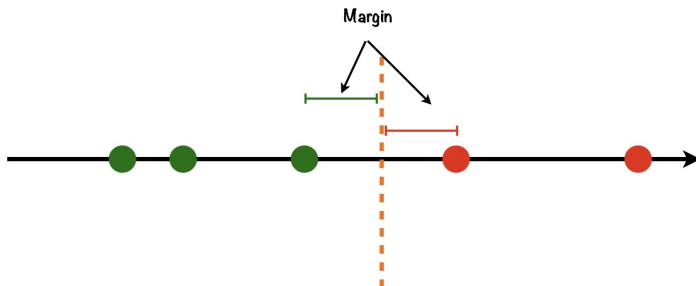
Margin is the minimum distance to the classification boundary



Classifying with Maximum Margin

Maximal Margin

We try to maximize the margin



Binary Classification

We now try to formulate everything concretely

Consider a binary classification problem with dataset

$$\mathbb{D} = \left\{ (\mathbf{x}_n, v_n) : \mathbf{x}_n \in \mathbb{R}^d \text{ and } v_n \in \{-1, 1\} \right\}$$

We want to use a linear classifier, i.e.,

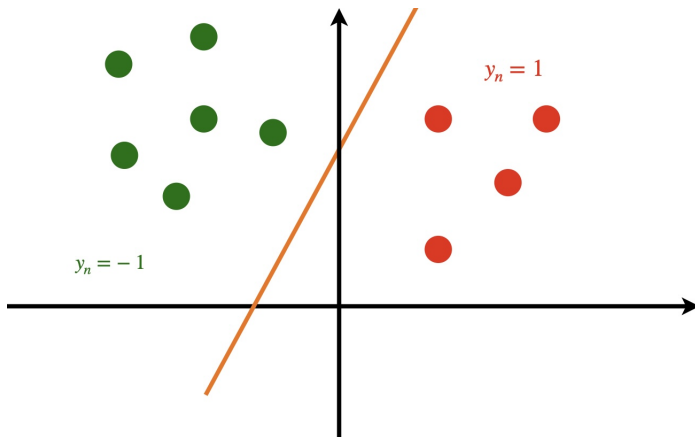
$$z_n = \mathbf{w}^\top \mathbf{x}_n \rightsquigarrow y_n = \begin{cases} 1 & z_n \geq 0 \\ -1 & z_n < 0 \end{cases}$$

A New Desire

We have maximal margin $\equiv$ confident classification

Binary Classification

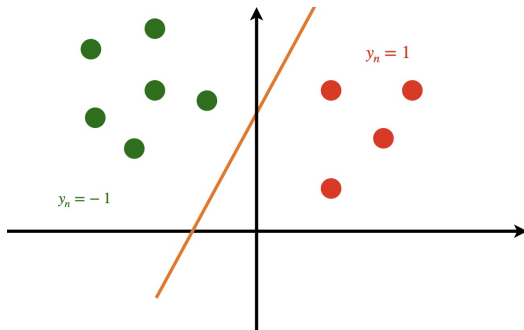
Recall that linear classifier is separation by a hyperplane



Binary Classification: *Error-Free Classification*

With perfect boundary, we have

$$y_n = \begin{cases} 1 & v_n = 1 \\ -1 & v_n = -1 \end{cases} \rightsquigarrow z_n = \begin{cases} > 0 & v_n = 1 \\ < 0 & v_n = -1 \end{cases} \rightsquigarrow v_n \mathbf{w}^T \mathbf{x}_n > 0$$



Binary Classification: *Forcing Maximal Margin*

In **basic linear classification**, we want zero error on training set, i.e.,

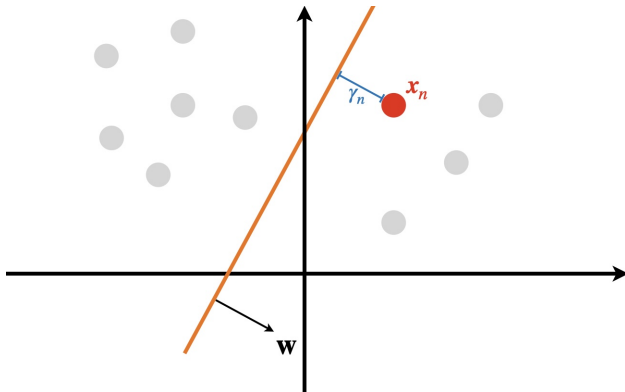
Find $\mathbf{w}$ such that $v_n \mathbf{w}^\top \mathbf{x}_n > 0$ for all n

If we want to have also **maximal margin**, we can write

$\max_{\mathbf{w}}$ **Margin** subject to $v_n \mathbf{w}^\top \mathbf{x}_n > 0$ for all n

? *How we can express the margin?*

Finding Margin: Positive Side



Note

w is orthogonal to the boundary

Finding Margin: Positive Side

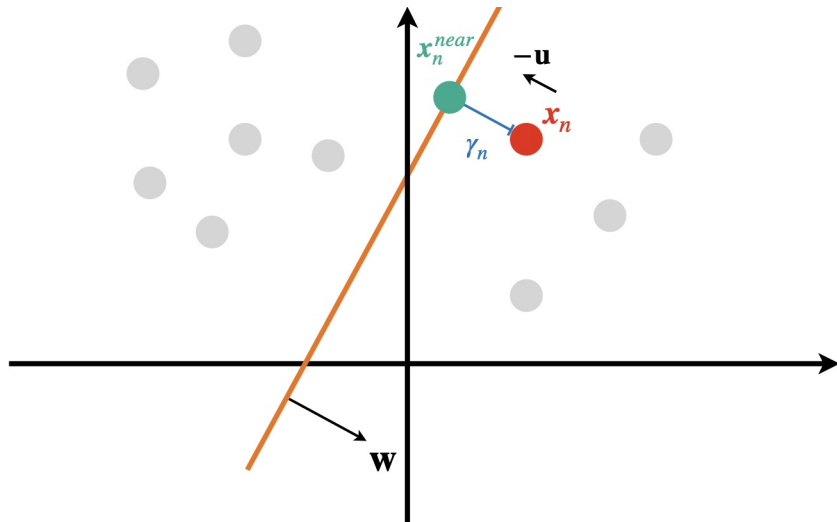
Unit vector orthogonal to boundary $\mathbf{w}^\top \mathbf{x} = 0$

$$\mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$$

Margin of point $\mathbf{x}_n$ is step γ_n to the nearest point on the boundary

$$\mathbf{x}_n^{near} = \mathbf{x}_n - \gamma_n \mathbf{u} = \textit{on the boundary}$$

Finding Margin: Positive Side



Finding Margin: Positive Side

Any point on the boundary satisfies $\mathbf{w}^\top \mathbf{x} = 0$

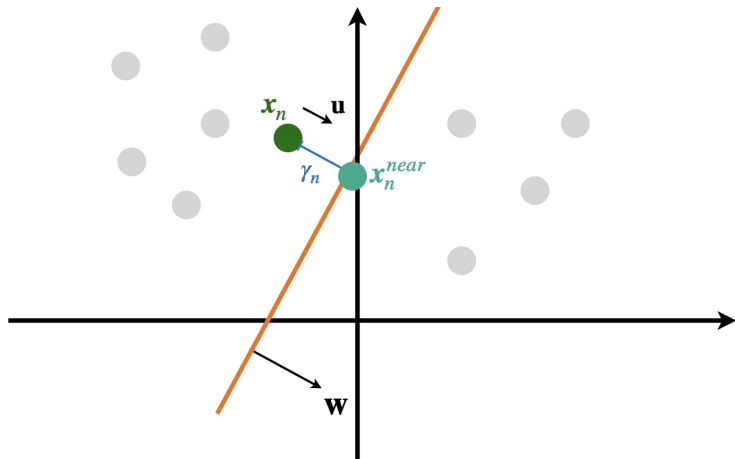
$$\mathbf{w}^\top \mathbf{x}_n^{near} = \mathbf{w}^\top (\mathbf{x}_n - \gamma_n \mathbf{u}) = 0$$

This concludes

$$\gamma_n = \frac{\mathbf{w}^\top \mathbf{x}_n}{\mathbf{w}^\top \mathbf{u}} = \frac{\mathbf{w}^\top \mathbf{x}_n}{\|\mathbf{w}\|}$$

Finding Margin: Negative Side

? What if we are on the other side, i.e., $v_n = -1$?



Finding Margin: Negative Side

When $v_n = -1$, we can write

$$\mathbf{w}^\top \mathbf{x}_n^{near} = \mathbf{w}^\top (\mathbf{x}_n + \gamma_n \mathbf{u}) = 0$$

This concludes

$$\gamma_n = -\frac{\mathbf{w}^\top \mathbf{x}_n}{\mathbf{w}^\top \mathbf{u}} = -\frac{\mathbf{w}^\top \mathbf{x}_n}{\|\mathbf{w}\|}$$

Finding Margin: Generic

We can say in general that for any $(\mathbf{x}_n, v_n)$, we have

$$\gamma_n = v_n \frac{\mathbf{w}^\top \mathbf{x}_n}{\|\mathbf{w}\|}$$

Margin

Margin is the shortest distance to the boundary

$$\mathcal{M} = \min_n v_n \frac{\mathbf{w}^\top \mathbf{x}_n}{\|\mathbf{w}\|} = v_{n^*} \frac{\mathbf{w}^\top \mathbf{x}_{n^*}}{\|\mathbf{w}\|}$$

$(\mathbf{x}_{n^}, v_{n^*})$ is the point closest to the boundary*

Finding Margin: Key Observation

? *Does it matter how large $\mathbf{w}$ is scaled?*

Let us scale $\mathbf{w}$ as $\mathbf{w} \leftarrow \alpha \mathbf{w}$, we have

$$\gamma_n = v_n \frac{\alpha \mathbf{w}^\top \mathbf{x}_n}{\alpha \|\mathbf{w}\|} = v_n \frac{\mathbf{w}^\top \mathbf{x}_n}{\|\mathbf{w}\|}$$

! *No!*

Conclusion

We can scale $\mathbf{w}$ as we wish!

Finding Margin

Let's scale $\mathbf{w}$ such that the closest point satisfies

$$v_{n^*} \mathbf{w}^\top \mathbf{x}_{n^*} = 1$$

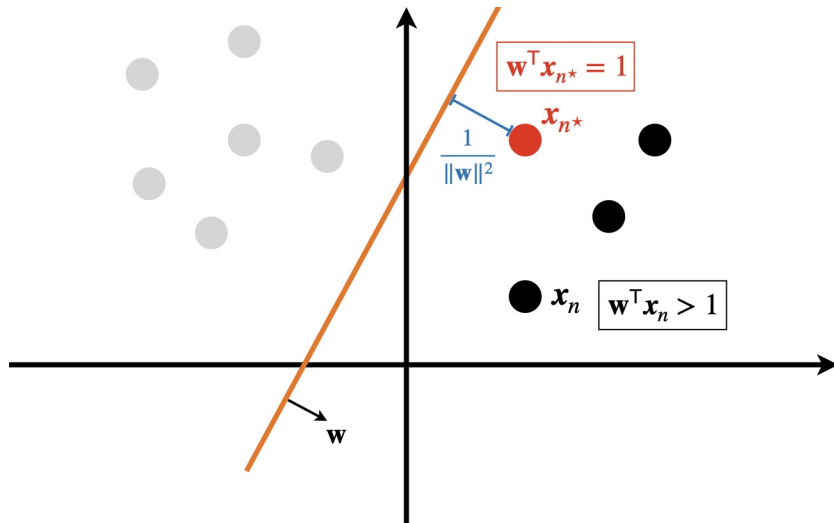
- *For the nearest points, we have*

$$\gamma_{n^*} = v_{n^*} \frac{\mathbf{w}^\top \mathbf{x}_{n^*}}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|} = \mathcal{M}$$

- *For the all points in the dataset, we have*

$$v_n \mathbf{w}^\top \mathbf{x}_n \geq 1$$

Finding Margin: *Visualization*



Training with Maximal Margin

Attention

Note that for all points in the dataset, we have

$$v_n \mathbf{w}^\top \mathbf{x}_n \geq 1 \rightsquigarrow v_n \mathbf{w}^\top \mathbf{x}_n > 0$$

which guarantees zero classification error

We wanted to find $\mathbf{w}$ as

$$\max_{\mathbf{w}} \text{Margin subject to } v_n \mathbf{w}^\top \mathbf{x}_n \geq 0 \text{ for all } n$$

We can do it alternatively as

$$\max_{\mathbf{w}} \mathcal{M} \text{ subject to } v_n \mathbf{w}^\top \mathbf{x}_n \geq 1 \text{ for all } n$$

Classifying with Maximal Margin

$$\max_{\mathbf{w}} \mathcal{M} \quad \text{subject to } v_n \mathbf{w}^T \mathbf{x}_n \geq 1 \text{ for all } n$$

This is written as

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|} \quad \text{subject to } v_n \mathbf{w}^T \mathbf{x}_n \geq 1 \text{ for all } n$$

or alternatively

$$\min_{\mathbf{w}} \|\mathbf{w}\|^2 \quad \text{subject to } v_n \mathbf{w}^T \mathbf{x}_n \geq 1 \text{ for all } n$$

Looking at SVC: Constrained Optimization

Support Vector Classifier

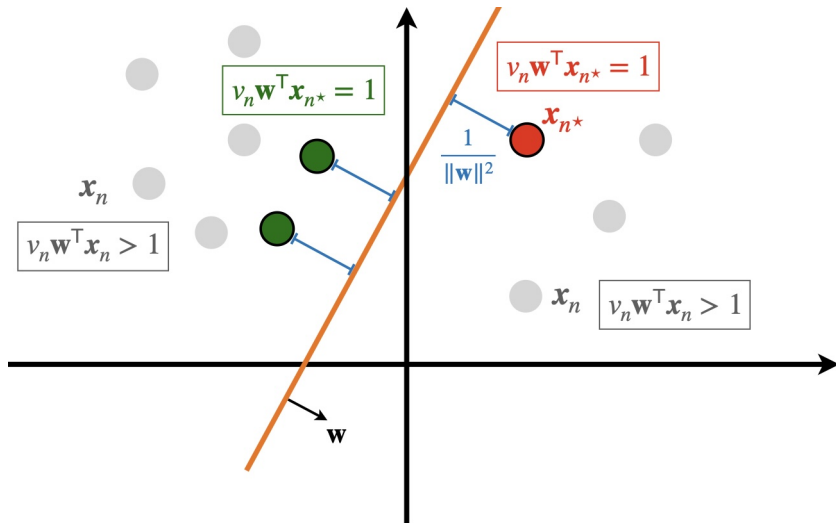
SVC with maximal margin finds $\mathbf{w}$ as

$$\min_{\mathbf{w}} \|\mathbf{w}\|^2 \quad \text{subject to } v_n \mathbf{w}^T \mathbf{x}_n \geq 1 \text{ for all } n$$

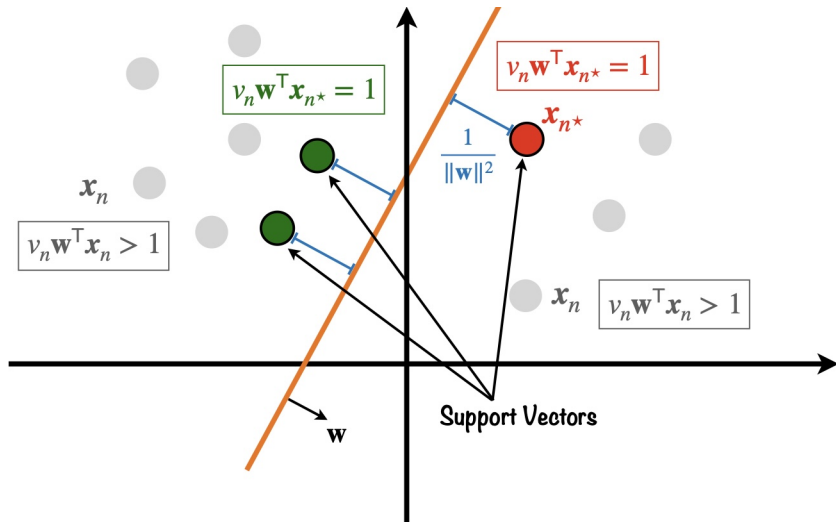
This is a constrained optimization

- We want to find minimum of $\|\mathbf{w}\|^2$
 - ↳ This is the objective function
 - ↳ With no constraint it's obvious: $\mathbf{w} = \mathbf{0}$
- But, we are constrained by $v_n \mathbf{w}^T \mathbf{x}_n \geq 1$ for all samples
 - ↳ With N samples, we have N constraints
 - ↳ Obvious solution is not valid, since $v_n \mathbf{0}^T \mathbf{x}_n = 0 \not\geq 1$

Looking at SVC: Visual Illustration



Looking at SVC: Visual Illustration



Optimization with Inequality Constraints

We need to develop some approach that lets us solve

$$\min_{\mathbf{w}} f(\mathbf{w}) \quad \text{subject to } g_n(\mathbf{w}) \leq 0 \text{ for all } n$$

In SVC, we have

$$\begin{aligned} f(\mathbf{w}) &= \|\mathbf{w}\|^2 \\ g_n(\mathbf{w}) &= 1 - v_n \mathbf{w}^T \mathbf{x}_n \end{aligned}$$

? *How we can solve this constrained optimization?*

Lagrange Dual Objective

We make the dual Lagrangian function: *let*

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_N \end{bmatrix}$$

be a vector of auxiliary variables; then, the dual Lagrangian is

$$\ell(\mathbf{w}, \boldsymbol{\lambda}) = f(\mathbf{w}) + \sum_{n=1}^N \lambda_n g_n(\mathbf{w})$$

where we have all Lagrange dual variables positive

$$\lambda_n \geq 0$$

Lagrange Dual Objective: *Key Property*

? Why do we define such a function?

Let's look at its maximum value over λ for an arbitrary $\mathbf{w}$

$$\ell(\mathbf{w}, \lambda) = f(\mathbf{w}) + \sum_{n=1}^N \lambda_n g_n(\mathbf{w})$$

and recall that $\lambda_n > 0$ for all n

- At *feasible points*: $g_n(\mathbf{w}) \leq 0$ for all n

$$\ell(\mathbf{w}, \lambda) = f(\mathbf{w}) + \sum_{n=1}^N \lambda_n g_n(\mathbf{w}) \leq f(\mathbf{w})$$

↳ we could thus say that

$$\max_{\lambda} \ell(\mathbf{w}, \lambda) = f(\mathbf{w})$$

Lagrange Dual Objective: *Key Property*

? Why do we define such a function?

Let's look at its maximum value over λ for an arbitrary $\mathbf{w}$

$$\ell(\mathbf{w}, \lambda) = f(\mathbf{w}) + \sum_{n=1}^N \lambda_n g_n(\mathbf{w})$$

and recall that $\lambda_n > 0$ for all n

- At *infeasible points*: $g_n(\mathbf{w}) > 0$ for all n

$$\ell(\mathbf{w}, \lambda) = f(\mathbf{w}) + \sum_{n=1}^N \lambda_n g_n(\mathbf{w}) \geq f(\mathbf{w})$$

↳ we could say that

$$\max_{\lambda} \ell(\mathbf{w}, \lambda) = +\infty$$

Lagrange Dual Objective: *Key Property*

? *Why do we define such a function?*

So, we have

$$\max_{\boldsymbol{\lambda} \geq 0} \ell(\mathbf{w}, \boldsymbol{\lambda}) = \begin{cases} f(\mathbf{w}) & g_n(\mathbf{w}) \leq 0 \\ +\infty & g_n(\mathbf{w}) > 0 \end{cases}$$

This means that

$$\min_{\mathbf{w}} \max_{\boldsymbol{\lambda} \geq 0} \ell(\mathbf{w}, \boldsymbol{\lambda}) = \min_{\mathbf{w}} f(\mathbf{w}) \quad \text{subject to } g_n(\mathbf{w}) \leq 0$$

! *We can solve this unconstrained problem instead!*

Solving Dual Problem

Primal Problem

This alternative form describes the primal problem

$$P = \min_{\mathbf{w}} \max_{\boldsymbol{\lambda} \geq 0} \ell(\mathbf{w}, \boldsymbol{\lambda})$$

*However, it is easier to solve the **dual problem***

Dual Problem

Dual problem solves the unconstrained minimization first

$$D = \max_{\boldsymbol{\lambda} \geq 0} \min_{\mathbf{w}} \ell(\mathbf{w}, \boldsymbol{\lambda})$$

Solving Dual Problem: *Strong Duality*

The key property is that the

dual value always bounds the primal from below, i.e., $D \leq P$

Strong Duality

Under Slater's conditions dual and primal values meet

- $f(\mathbf{w})$ is convex
- $g_n(\mathbf{w})$ are all convex
- There is at least one $\mathbf{w}_0$ such that $g_n(\mathbf{w}_0) < 0$ for all n

Solving Dual Problem: *KKT Conditions*

? *Say we have strong duality; then, how to find the solution?*

We need to find the point that satisfies KKT conditions

- *to be a stationary point*

$$\nabla_{\mathbf{w}} \ell(\mathbf{w}^*, \boldsymbol{\lambda}^*) = \mathbf{0}$$

- *to be feasible*

$$g_n(\mathbf{w}^*) \leq 0$$

$$\lambda_n^* \geq 0$$

- *to satisfy supplementary slackness*

$$\lambda_n^* g_n(\mathbf{w}^*) = 0$$

Solving Dual Problem: KKT Conditions

- ! The key point is supplementary slackness

$$\lambda_n^* g_n(\mathbf{w}^*) = 0$$

- If the optimal point is on boundary; then, the dual variable is *active*

$$g_n(\mathbf{w}^*) = 0 \rightsquigarrow \lambda_n^* > 0$$

- But if it's not on the boundary; then, dual variable is *inactive*

$$g_n(\mathbf{w}^*) < 0 \rightsquigarrow \lambda_n^* = 0$$

Example: Minimizing Paraboloid I

Let's find the solution to

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq 1$$

Before, we start

- without constraint the optimal point is at $\mathbf{w}^* = \mathbf{0}$
- it's already in the feasible region, since $w_1^* = 0 \leq 1$

Example: Minimizing Paraboloid I

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq 1$$

Slater's conditions hold, so we find dual objective

$$\ell(\mathbf{w}, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 - 1)$$

First, we find the stationary points

$$\nabla \ell(\mathbf{w}^*, \lambda^*) = \begin{bmatrix} 2w_1^* + \lambda \\ 2w_2^* \end{bmatrix} = \mathbf{0} \rightsquigarrow \begin{bmatrix} w_1^* \\ w_2^* \end{bmatrix} = \begin{bmatrix} -\lambda^*/2 \\ 0 \end{bmatrix}$$

Example: Minimizing Paraboloid I

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq 1$$

Slater's conditions hold, so we find dual objective

$$\ell(\mathbf{w}, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 - 1)$$

Next, we check the feasibility

$$w_1^* = -\frac{\lambda^*}{2} \leq 1 \rightsquigarrow \lambda^* \geq -2 \quad (\text{Primal})$$

$$\lambda^* \geq 0 \quad (\text{Dual})$$

so, we should have $\lambda^* \geq 0$

Example: Minimizing Paraboloid I

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq 1$$

Slater's conditions hold, so we find dual objective

$$\ell(\mathbf{w}, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 - 1)$$

Finally, we check the supplementary slackness

$$\lambda^*(w_1^* - 1) = 0 \rightsquigarrow -\lambda^* \left(1 + \frac{\lambda^*}{2}\right) = 0$$

Since we look for $\lambda^* \geq 0$, the only solution is

$$\lambda^* = 0$$

Example: Minimizing Paraboloid I

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq 1$$

The dual solution is

$$\begin{bmatrix} w_1^* \\ w_2^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and $\lambda^* = 0$ which says that the dual variable is *inactive*

This makes sense, since the constraint *does not really impact!*

Example: Minimizing Paraboloid II

Now, let's solve this problem

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq -1$$

Before, we start

- without constraint the optimal point is at $\mathbf{w}^* = \mathbf{0}$
- it's **infeasible**, since $w_1^* = 0 \not\leq -1$

For this simple example, we can easily see that the solution is on boundary then

Example: Minimizing Paraboloid II

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq -1$$

Slater's conditions hold, so we find dual objective

$$\ell(\mathbf{w}, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 + 1)$$

First, we find the stationary points

$$\nabla \ell(\mathbf{w}^*, \lambda^*) = \begin{bmatrix} 2w_1^* + \lambda \\ 2w_2^* \end{bmatrix} = \mathbf{0} \rightsquigarrow \begin{bmatrix} w_1^* \\ w_2^* \end{bmatrix} = \begin{bmatrix} -\lambda^*/2 \\ 0 \end{bmatrix}$$

Example: Minimizing Paraboloid II

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq -1$$

Slater's conditions hold, so we find dual objective

$$\ell(\mathbf{w}, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 + 1)$$

Next, we check the feasibility

$$w_1^* = -\frac{\lambda^*}{2} \leq -1 \rightsquigarrow \lambda^* \geq 2 \quad (\text{Primal})$$

$$\lambda^* \geq 0 \quad (\text{Dual})$$

so, we should have $\lambda^* \geq 2 \rightsquigarrow$ the dual variable cannot be **inactive!**

Example: Minimizing Paraboloid II

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq -1$$

Slater's conditions hold, so we find dual objective

$$\ell(\mathbf{w}, \lambda) = w_1^2 + w_2^2 + \lambda(w_1 + 1)$$

Finally, we check the supplementary slackness

$$\lambda^*(w_1^* + 1) = 0 \rightsquigarrow \lambda^* \left(1 - \frac{\lambda^*}{2}\right) = 0$$

Since we look for $\lambda^* \geq 2$, the only solution is

$$\lambda^* = 2$$

Example: Minimizing Paraboloid II

Now, we solve it using the Lagrange multipliers method

$$\min w_1^2 + w_2^2 \text{ subject to } w_1 \leq -1$$

The dual solution is

$$\begin{bmatrix} w_1^* \\ w_2^* \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

and $\lambda^* = 2$ which says that the solution is *on the boundary*

This makes sense, since the constraint *is now actively impacting!*

Further Read

- Bishop

- ↳ Chapter 7: *Section 7.1*

SVC

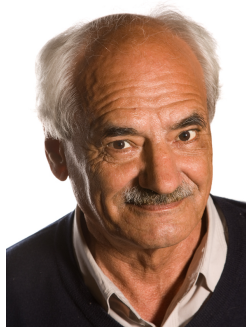
- ESL

- ↳ Chapter 12: *Section 12.1 - 12.2*

SVC

Acknowledgment

Vladimir Vapnik and Alexey Chervonenkis



Take a look at [VC Theory](#)